

Reasoning Over Earth Engine.

An AI Agent for Kenya's Climate Risk.

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THE PROBLEM.

Climate data is everywhere. Answers aren't.



Seasonal Impact

Floods and droughts hit Kenyan counties every season.



Complexity Barrier

The satellite data exists - but reading it needs experts.



Operational Delay

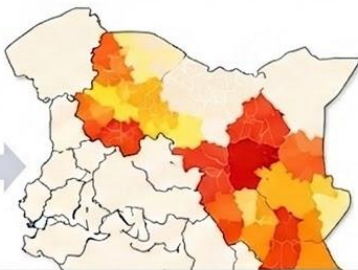
Decisions are slow when every question means hours of GIS work.

OUR SOLUTION

Ask. Get a grounded answer.

"Which counties face the highest flood risk, and who's exposed?"

Gemini



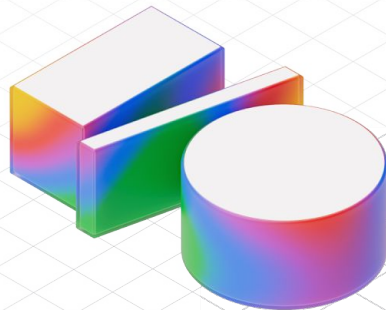
Kenya Flood Risk

- Highest Risk
- High Risk
- Moderate Risk

1. Nairobi
2. Garrisa
3. Turkana
4. Homa Bay

Insightful Analysis

Over 50,000 people at high flood risk in Nairobi County.



WHAT'S NEW (Google I/O 2026)

Built on Google I/O 2026

We used directly:

- **Gemini 3.5 Flash** - the new model; fast enough for live agent loops (*announcement #1*).
- **AGENTS.md** - New markdown formats Google announced for defining an agent (*#61*).
- **SKILL.md** - Markdown for defining Skills (*#61*).

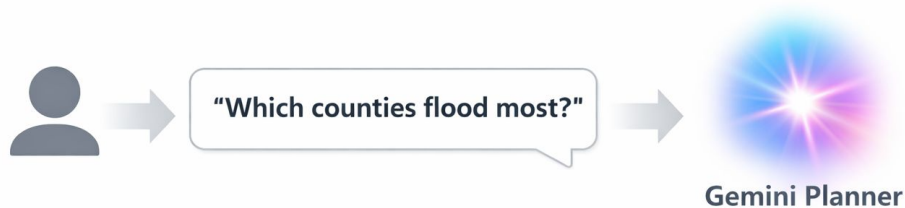
Inspired by Google I/O 2026

Built ourselves:

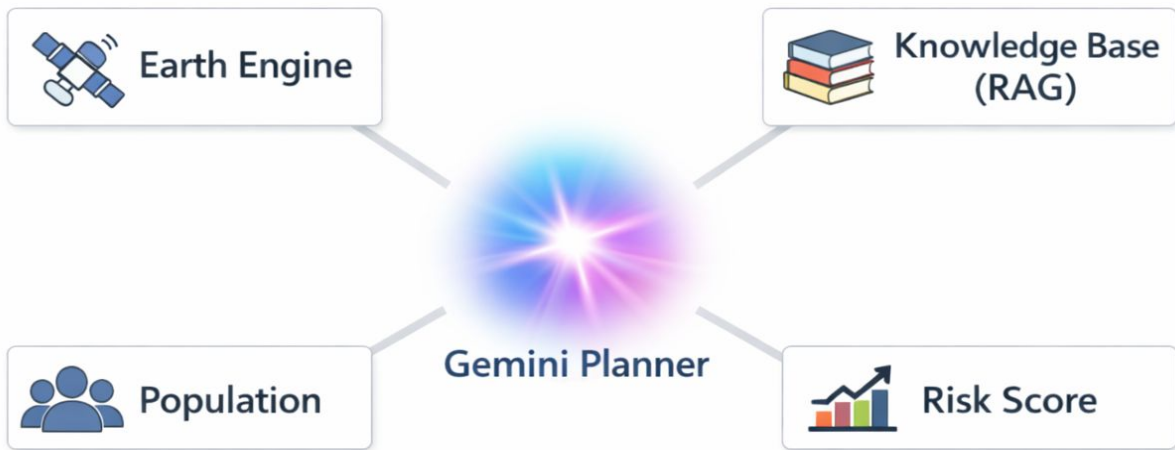
- **Agentic tool-calling + sub-agents** - expert tools echoing agent-harness primitives
- **Managed-Agents-style orchestration** - inspired by Managed Agents (*#60*).
- **Grounded retrieval (RAG)** - answers citing real Kenyan documents or sources.

METHODOLOGY.

It starts with a question...

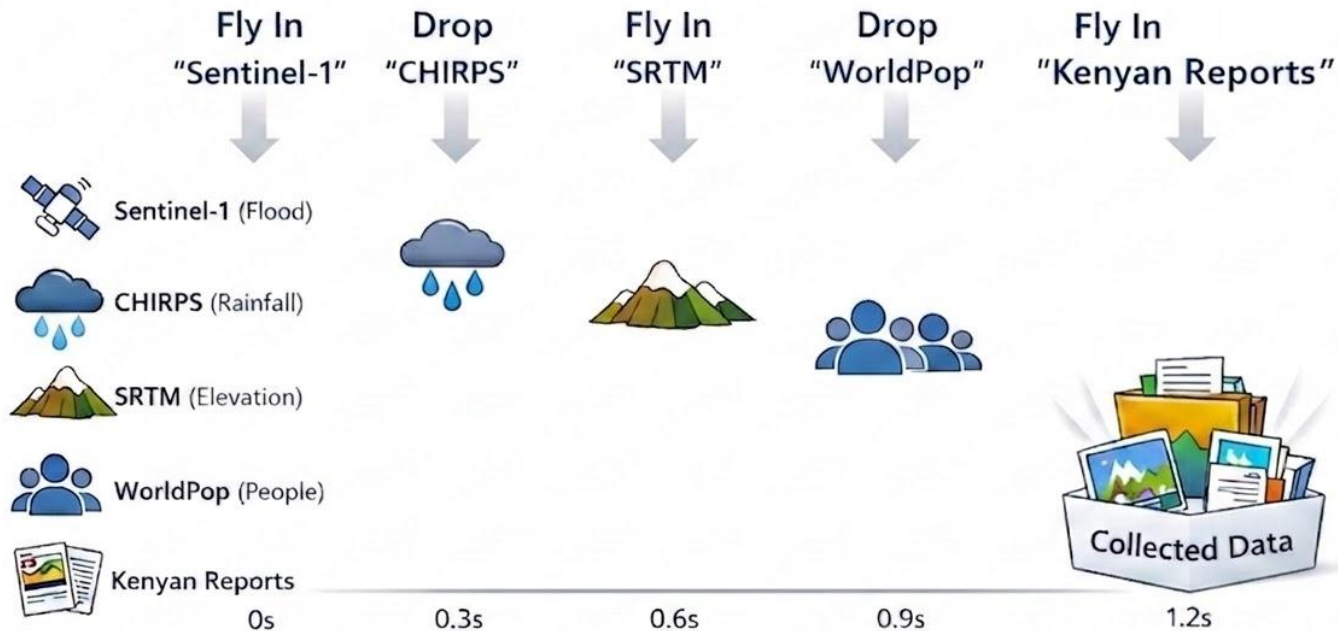


Gemini picks the right tools



The agent decides what it needs.

Pulling real-world signals



A transparent risk score – not a black box



From Data to Decision



Ranked Counties



Interactive Map

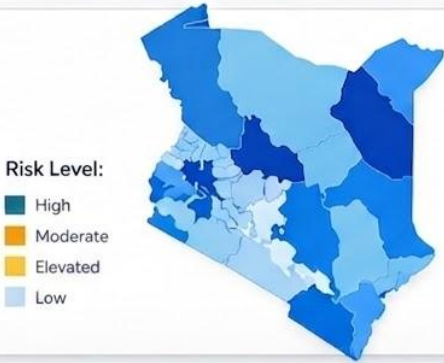


Recommendations
+ Cited Reasoning

Top Counties by Risk

Rank	County	Risk	Exposed Population
1.	Turkana	High	250,000
2.	Garissa	Elevated	180,000
3.	Kilifi	Elevated	210,000
4.	Wajir	Moderate	160,000
5.	Mandera	Guarded	140,000

Kenya Risk Map



Turkana is at high risk due to severe drought and limited resources (UNDP, WFP).
Over 250,000 people are exposed. Addressing this is a top priority.

94% Confidence ✓

The *Full* Picture





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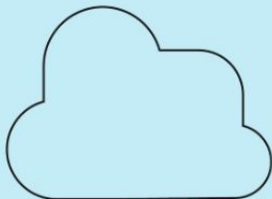


CODELAB:

Type This Link:

<https://shorturl.at/nVsNm> - Github

<https://shorturl.at/1qb2D> - Colab



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